TinyJ Assignment 2*

The focus of this assignment is syntax translation from TinyJ to virtual machine instructions. Your compiler does all of the following whenever its input is a syntactically valid TinyJ source file:

1. It checks that declarations and uses of identifiers in the source file are consistent with Java's scope rules.

2. As long as no errors are detected in the source file, **it translates the source file into a sequence of instructions for a stack-based virtual machine whose instruction set is given in Table 1**. At the same time, it writes to the output file an "enhanced parse tree" of the source file; this shows the static address or the stack frame offset that the compiler has allocated to each integer or array reference variable, the start address of the code generated for each method, and the time at which each instruction is generated.

3. If no errors are detected in the source file then a list of the instructions generated is also written to the output file.

A specification of the effects of executing these instructions is given on pp. 5 – 6 of the document "Asn-2-Memory-allocation-VM-instruction-set-and-hints.pdf".

1 How to Setup Assignment 2

You should complete the setup on ***mars*** and test all the assignments on ***mars***. To install assignment 2, do the following steps:

1. Login to ***mars*** (https://www.cs.qc.cuny.edu/mars/).

2. Change to the TinyJ directory (optional): `cd TinyJ`

3. Enter the following on ***mars***: `/home/faculty/ctsai/TJ2setup`

4. Wait for the line "TJ2setup done" to appear on the screen.

5. If you choose to do the assignment on your PC or Mac, you manually execute commands in TJ2setup in your working directory in the powershell window:
```
jar xvf TJasn.jar
javac -cp .  TJasn/TJ.java
javac -cp .  TJasn/virtualMachine/*.java
```

2 How to do TinyJ Assignment 2

`Parser.java` from assignment 1 becomes `TJasn/ParserAndTranslator.java`, which is the only file you should fix. In this file, each /* ???????? */ comment must be replaced with appropriate code. You may first copy over code from a **backup copy** of Parser.java from assignment 1 and then make changes to generate appropriate TinyJ virtual machine instructions via translation. To recompile `TJasn/ParserAndTranslator.java`, do:
```
javac -cp .  TJasn/ParserAndTranslator.java
```

*Courtesy of Prof. Tatyung Kong

Operation	**Operand 1**	**Operand 2**
STOP		
PUSHNUM	⟨ integer ⟩	
PUSHSTATADDR	⟨ address of static variable ⟩	
PUSHLOCADDR	⟨ local variable or parameter's stackframe offset⟩	
LOADFROMADDR		
SAVETOADDR		
WRITELNOP		
WRITEINT		
WRITESTRING	⟨address of first char⟩	⟨address of last char⟩
READINT		
CHANGESIGN		
NOT		
ADD		
SUB		
MUL		
DIV		
MOD		
AND		
OR		
EQ		
LT		
GT		
NE		
GE		
LE		
JUMP	⟨address of target instruction⟩	
JUMPONFALSE	⟨address of target instruction⟩	
CALLSTATMETHOD	⟨address of method's first instruction⟩	
INITSTKFRM	⟨no. of locations needed for local vars declared in the method's body⟩	
RETURN	⟨no. of parameters the method has⟩	
HEAPALLOC		
ADDTOPTR		
PASSPARAM		
DISCARDVALUE		
NOP		

Table 1: Virtual machine instructions

3 How to test Your Parser

If your program cannot be compiled error-free on ***mars***, you will receive no credit. To run your program on any TinyJ source file enter the following command:

`java -cp .  TJasn.TJ TinyJ-source-file-name output-file-name`

Example: `java -cp .  TJasn.TJ CS316ex12.java 12.out`

For this assignment, answers for questions during execution:

- Want debugging stop or post-execution dump? (y/n) [Ans: y]
- Enter MINIMUM no. of instructions to execute before debugging stop. ... [Ans: 0]
- Stop after executing what instruction? (e.g., PUSHNUM) ... [Ans: *]

You can check the correctness of your work with Prof. Kong's solution by doing the following:

`java -cp TJsolclasses:.  TJasn.TJ TinyJ-source-file-name output-file-name` [on ***mars*** and Mac]

`java -cp "TJsolclasses:." TJasn.TJ TinyJ-source-file-name output-file-name` [on PC]

When the TinyJ source file is a correct TinyJ program, the output files produced by yours and the solution program must contain the same "Instructions Generated:" lists (near the end, just above the "Data memory dump"). But the output files need not be identical.

4 How to Submit

To submit:

1. Leave your final version of `ParserAndTranslator.java` on ***mars*** in your `TJasn` directory, before midnight on the due date.
2. Make a backup copy for Assignment 2: `cp -r TinyJ TinyJ-assign2-bak`
3. Grant the instructor the access permission, if not done for Assignment 1:
   ```
   chmod 711 $HOME
   cd $HOME
   setfacl -R -m u:ctsai:rwX TinyJ/
   ```
4. Only one submission per group on Brightspace. Students should stay in the same group as for Assignment 1.
5. Each submission contains the absolute directory of your working directory and name(s) of student(s).
6. Do not touch/open `ParserAndTranslator.java` again after the due date, so not to change the time stamp.